

Superstore — Pandas Practice

2-day practice set • Try without solutions first • Focus on thinking, not memorizing syntax

Dataset: Superstore sales CSV

How to use this: Spend around 10–15 minutes on a question. If you're stuck, first write down what the final result should look like and what operations you think might be needed. Then check documentation or a solution and redo it yourself.

■ Level 1 — Warm-up

1. Find all orders where **Sales > 500**.
2. Find all orders where **Category = Technology** and **Profit > 100**.
3. Find the **10 orders with the highest Sales**.
4. Display only **Customer Name, Product Name, Sales, and Profit** for orders placed in 2017.
5. Create a column called **Profit Margin** using **Profit / Sales**. Find the 10 orders with the highest profit margin.

■ Level 2 — GroupBy & Aggregation

1. Calculate total **Sales**, total **Profit**, and average **Discount** for each Category.
2. Calculate total Sales and Profit for each Region.
3. Find the **top 10 customers by total Sales**.
4. Find the **top 5 sub-categories by total Profit**.
5. Calculate the **average Sales per order** for each Segment.

■ Level 3 — Start Thinking

1. Find the **top 3 products by Sales within each Category**.
2. Find the product with the **highest total Profit** (not the highest individual-order Profit).
3. Find products where **Total Sales > 0** but **Total Profit < 0**.
4. Find the **10 customers who generated the highest total Profit**.
5. For each Category, calculate what **percentage of the company's total Sales** comes from that category.

■ Level 4 — Interview-ish

1. For every customer, find their **first order date**, **most recent order date**, and **number of days between them**.
2. Find customers who placed **more than one order**.
3. Calculate total Sales for every month, then calculate the **month-over-month Sales change**.
4. Calculate **cumulative Sales over time**.

5. For every Region, rank the Categories by total Sales.

■ Bonus — Analyst Thinking

1. Which customers have spent **more than the average customer**?
2. Which products have **high sales but relatively low profit**?
3. For each Region, find the **month with the highest Sales**.
4. Find the **top 3 customers in each Region** based on total Sales.
5. Find the **percentage of orders that resulted in a loss**.

Suggested 2-Day Plan

Day	Focus	Questions
Day 1	Basics + GroupBy	Level 1 + Level 2 (1–10)
Day 2	Interview-style thinking	Level 3 + Level 4 (11–20)
Bonus	Problem decomposition	Bonus 21–25; focus on approach first

Important: Don't try to finish all 25. The goal is to recognize patterns, translate a question into steps, and become comfortable solving unfamiliar data problems.

After this: Move from topic-wise practice to mixed questions, then use StrataScratch for more interview-style problem decomposition.